

Virtual PIN: Biometric Encryption Using Coding Theory

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Abstract: A new biometric system is proposed which replaces the template match known from presently used biometric systems by a more powerful approach that uses error tolerant coding. This enables the system for new applications and enhances compatibility to non-biometric systems which are in use now.

1. Introduction

All biometric systems that are in use today rely on matches against templates previously stored. This means that before the system can be used in a first step, the enrolment, a sample of the biometric trait is acquired and data extracted from that trait is saved as a biometric template somewhere. Later, during authentication, a new sample of the biometric trait is acquired and the data extracted from that trait is compared with the reference template. The result of that comparison is a match score, a numerical measure of the similarity of the two traits acquired at enrolment and verification.

This score may be compared to a threshold, giving ultimately a binary answer whether the similarity was sufficient or not.

No further use is made of the information contained in the biometric trait.

Here, a new system is proposed where the information contained in the biometric trait is not only used to obtain the match score but that goes one step further to extract data in a repeatable way from the trait. That data may be used for different purposes, including PIN authentication or cryptographic applications.

The data contained in the biometric trait will be called virtual PIN here, because the use of this data may be seen as an analogue to the use of a PIN entered using a key pad.

Since no two realisations of a biometric trait are identical, it is necessary to correct the difference between the data acquired in the enrolment step and the verification step. This correction replaces the matching step of present biometric systems. In order to perform that correction low density parity check codes (LDPC) are used.

This paper is organised as follows:

First, an overview of the complete system is given. Then, details on the data extraction algorithm and the error correction algorithm are given, followed by preliminary results of the proof-of-concept implementation and an outlook towards a workable implementation.

2. Overview of the implementation

Like any other biometric system, the virtual PIN concept knows two different modes of operation known as enrolment and verification.

The input to the enrolment consists of

- a biometric trait, for example a fingerprint image acquired with a live fingerprint sensor
- a secret to be protected by the biometrics, in other words, the virtual PIN
- a generator matrix that defines an error correcting code.

The information contained in the biometric trait is extracted in form of a feature vector. The defining property of the feature vector is that a given piece of information is always stored at the same position of the feature vector. Therefore the feature vectors extracted from two matching fingerprint images are similar even if the fingers have been put on the sensor at two very different positions or in two very different orientations.

The secret may come from some outside source or may be randomly chosen.

The generator matrix G for the LDPC code is taken from well understood results of coding theory [MN99].

Enrolment

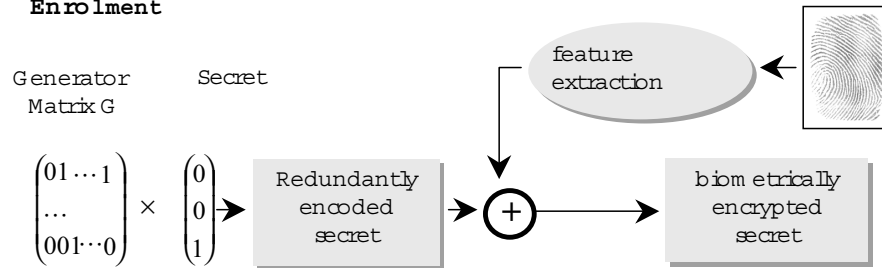


Figure 1: An example of the enrolment process consisting of the following steps:

1. feature extraction algorithm applied to the fingerprint image,
2. redundant encoding of the secret,
3. combination of the feature vector that results from the feature extraction and the encoded secret to the biometrically encrypted secret. In this example the combination is a XOR operation.

The secret is multiplied by the generator matrix to obtain the redundantly encoded secret. Then the secret is XORed with the feature vector to obtain the biometrically encrypted secret. The biometrically encrypted secret is then stored on a data carrier, for example a smart card. It is not possible to gain the secret from this data without having the biometric data or to obtain the biometric data from without having the secret (Figure 1).

From a cryptographic point of view that requires that there are no correlations in the biometric data. This can, for example, be achieved using a Karhunen-Loève transformation [Ka46][Lo55].

At authentication a similar procedure is used. A new image is acquired, feature extraction performed on that image and the new feature vector is XORed with the biometrically encrypted secret. Because the two feature vectors most probably are different, the result from that operation will be different from the redundantly encoded secret calculated in the enrolment step. This can be interpreted as the transmission of the encoded secret through a lossy channel. Since the secret had been redundantly encoded before, a decoder as known from error tolerant coding theory may now be used to fix these errors (Figure2).

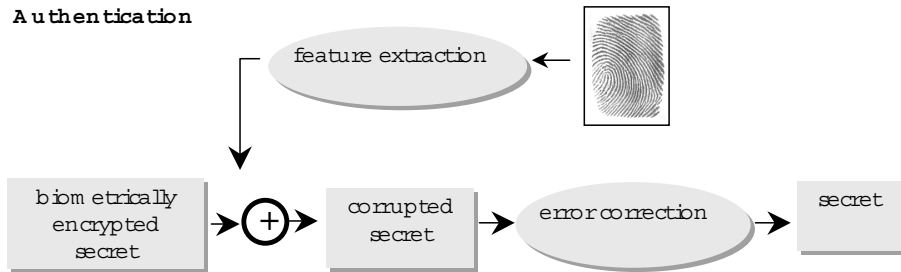


Figure 2: An example of the Authentication process consisting of the following steps:

1. feature extraction,
2. combination of the feature vector and the biometrically encrypted secret to a corrupted secret using a XOR operation,
3. elimination of the errors using a decoder as known from error tolerant coding theory.

3. Feature extraction using Gabor filters

The most critical part of the work presented here is the extraction of the biometric information in form of a feature vector from the realisation of a biometric trait, in our case a fingerprint image. Because of the special requirements of the error correction the feature vectors used here are different from the biometric features used in other biometric systems.

Every entry in the feature vector must always correspond to the same piece of biometric information. This would not be the case in the common example of minutiae lists. The lists simply contain an enumeration of ridge endings and bifurcations. The position of a particular minutia in that list depends on the orientation of the fingerprints and the number minutiae detected elsewhere [OG55].

Because of this dependence such a minutiae enumeration is not suitable as input to the error correcting codes.

One way to implement such a feature vector is to define a coordinate system that is attached to the fingerprint image and not to the fingerprint sensor. In order to do so one needs a reference point and an orientation of the fingerprint image.

Figure 3 shows how the core orientation of a “right loop” fingerprint is detected. The core is essentially the maximum of the ridge curvature, and the orientation is determined in the ring around the core shown in inverted colours. At some angles above the core the ridges are tangential to the ring, while at other angles below the core the ridges are radial. The middle of the sector where the ridges are radial gives the orientation of the fingerprint image.

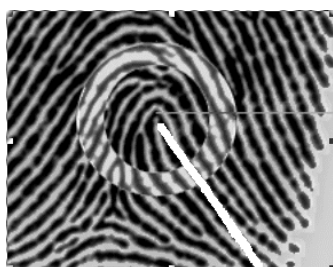


Figure 3: Orientation detection for a core. The orientation is determined using the pixels of the ring shown with inverted grey values. The orientation is indicated by the thick white line. The algorithm uses the fact that the orientation of the ridges points towards the core where the white line crosses the ring, while it is tangential in the upper left side of the ring.

Other similar algorithms are used for arches and whorls. In the case of an arch there are two sectors of radial ridges to the right and to the left of the core, and in the case of a whorl there are ridges bending upwards below the core that may be used to determine the orientation. Different fingerprint types are shown in Figure 4.



Figure 4: Arch, left loop and whorl

Following the literature [JPHP00], Gabor filter responses have been used to implement the feature vector extraction. Gabor filters are sensitive to any pattern that is periodic in one particular direction. This direction is an input parameter to the Gabor filters. The definition of the Gabor filter in terms of their convolution kernel K_{ij} is given in Figure 5.

The fingerprint is segmented into small regions as shown in Figure 6. These segments are large enough to contain at least one complete ridge and small enough to contain mostly parallel ridges. For each segment several Gabor filter responses are calculated with different orientations.

The Gabor filter response at a given point depends only on input points in the proximity of the output points, since the Gabor filter kernel contains a Gaussian of the distance of input and output coordinate. If the width of the Gaussian in the kernel corresponds approximately to the ridge distance of the fingerprint image the Gabor filter is very sensitive to fingerprint ridges.

Because of the Gaussian envelope, the Gabor filter acts as an orientated smoothing operation. Remaining random noise will thus be removed by the Gabor filters. An example of a Gabor filtered fingerprint image is shown in Figure 6.

Given the definitions
 $x_{ij} = -i \sin \varphi - j \cos \varphi$
 $y_{ij} = i \cos \varphi - j \sin \varphi$
 $k_{ij} = \cos(2\pi f x_{ij}) \exp(-1/2(x_{ij}^2/\sigma_x^2 + y_{ij}^2/\sigma_y^2))$
the elements of the Kernel are
 $K_{ij} = \sum_{k,l} k_{ij} / s^2$

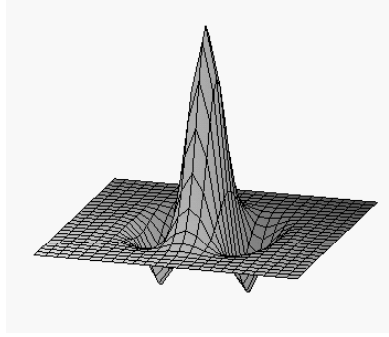


Figure 5: Gabor Kernel. The indices i and j run symmetrically around zero from $-(s-1)/2$ to $(s-1)/2$. Therefore, the size s of the kernel has to be an odd. The angle φ is the direction in which the Gabor filter „sees“ variations in the image to be filtered. The parameters σ_x and σ_y give the width of the Gaussian envelope of the filter and act as a smoothing parameter, the frequency f should be the inverse of the ridge distance in order to enhance ridges.

For each segment as shown in Figure 6 the variance of these Gabor filter responses is calculated. These variances then serves as feature vector entry.

The coordinates of the segments are given in the coordinate system attached to the fingerprint. Therefore, errors in the coordinate system translate to errors in the segmentation and ultimately to false non-matches.

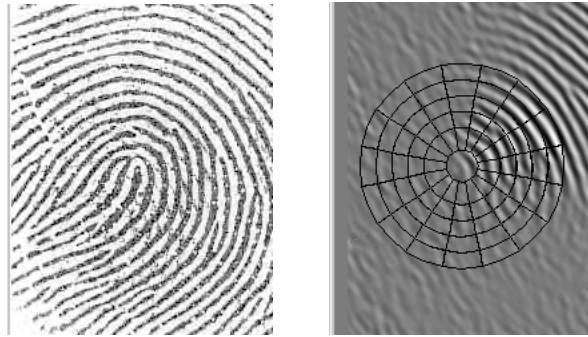


Figure 6: Original image and corresponding Gabor filter response. The waves in the upper right sector of the Gabor response indicate that in that sector the ridges had approximately the direction orthogonal to the Gabor filter. The orientation of the Gabor filter is approximately in the north-east direction.

To assess the discriminative power of the system, the difference of two feature vectors extracted from a pair of matching fingerprint images are plotted together with the difference of two feature vectors extracted from a pair of non-matching fingerprint images (Figure 7). The comparison shows that the differences between two matching feature vectors are much smaller than the differences between two non-matching feature vectors.

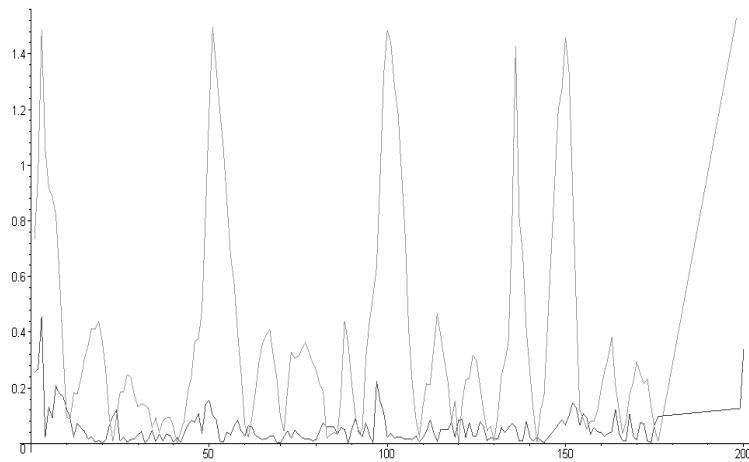


Figure 7: Comparison of the distances between two matching and two non-matching feature vectors. The graph shows the distances between the entries of two feature vectors corresponding to the same finger (green respectively black) or different fingers (red respectively grey). The distances corresponding to different fingers are significantly larger than the distances corresponding to the same finger. This shows that the feature extraction produces meaningful data. The horizontal axis shows the feature vector index, the vertical axis the difference of the two corresponding feature vector entries in angle differences.

The feature vectors consists of floating point numbers. In order to feed them into the error correcting codes, it is necessary to digitise them by approximating them with integer numbers and representing these integer numbers by Gray codes.

4. Error correction using Low Density Parity Check Codes

Low Density Parity Check Codes (LDPC) codes have been introduced in the 1960s by R. J. Gallager [Ga63]. LDPC codes are linear block codes. Actually LDPC codes constitute a large family of codes.

A linear code may be described in terms of a generator matrix $\mathbf{G}$ or in terms of a parity check matrix $\mathbf{H}$. The codewords corresponding to the Generator matrix $\mathbf{G}$ satisfy the condition. $\mathbf{H}\mathbf{x} = \mathbf{0}$.

If the parity check matrix $\mathbf{H}$ satisfies the conditions given below, the code corresponding to $\mathbf{H}$ is called a LDPC code.

- The number of non-zero entries in each column of $\mathbf{H}$ is a fixed number α .
- The number of ones in each row of $\mathbf{H}$ is a fixed number β .
- Both numbers are small compared to the corresponding matrix dimensions.

Typical parameters for regular LDPC codes are $(\alpha, \beta) = (3, 6)$.

Iterative decoding of LDPC codes can be implemented based on a message-passing algorithm called belief propagation or probability propagation algorithm [MN99].

Given some input $\mathbf{x}$ to be decoded, the vector $\mathbf{z} = \mathbf{H}\mathbf{x} \bmod 2$ is calculated, which is called the syndrome of $\mathbf{x}$. The message passing algorithm now determines modifications to the input vector $\mathbf{x}$ and $\mathbf{z}$ is calculated again. This step is iteratively repeated until $\mathbf{z}$ is zero or a maximum number of iterations is exceeded. In the last case, the decoding has failed, while in the first case the decoding may have reached the correct codeword or a wrong codeword. If the error of $\mathbf{x}$ is sufficiently small, the decoding is always successful. Effectively, the decoder finds the most probable vector $\mathbf{x}'$ such that $\mathbf{H}\mathbf{x}' \bmod 2 = \mathbf{0}$, given the input vector $\mathbf{x}$.

The advantage of LDPC codes over other block codes is that LDPC codes can have very large block sizes and relatively high error rates. Since our feature vectors contain many entries and have a high error rate, these properties are necessary in the context of the virtual PIN.

Furthermore, the decoder described here can be implemented efficiently.

5. Results and Outlook

Using the LDPC codes we obtain an equal error rate of 12% and approximately 7% with an conventional template matcher.

This shows that the codes are not yet adapted optimally to the feature vectors. A large potential for optimisations is expected here. In particular in the future the statistics of the feature vectors will be used for optimising the error correcting codes and the digitising of the feature vectors.

The matching rates obtained from the proof-of-concept prototype show that the virtual PIN is possible but are not good enough for a real-world-system. Our experience and the results of the FVC2002 [FVC02] fingerprint verification contest show good match rates are only possible with hybrid matchers using minutiae and pattern information like the Gabor filter responses. Therefore the next step will be to include minutiae information in the feature vector.

Completely different biometric systems, in particular iris biometrics seem also very promising. When using iris biometrics the problem of finding a reference frame in the image is much easier and the low false match rates of iris biometrics indicate that a large amount of information may be extracted from an iris.

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